

Hajee Mohammad Danesh Science and Technology University
MSc in CSE

Dept. of Computer Science

Course: Software Project Management

Course Code: CSE 531

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Reference Books: 1. Introduction to Software Project Management by Adolfo Villafiorita
 2. Software Engineering by Ian Sommerville

Making IT Perfect: Managing People and Organizing Communication

- Two primary resources for creating quality software:
 - People's intellect
 - People's ability
- While techniques and processes help control the software production environment, it is ultimately people who:
 - Transform ideas into requirements
 - Convert requirements into software
- Effective management of people and teams is crucial in software development projects.

Human resource management in a project involves four key steps:

- **Define Staff Requirements:** Identify the number of personnel needed and their required skills and competencies.
- **Select Staff:** Choose project participants, either from within the organization or by hiring new resources.
- **Manage Staff:** Focus on building a cohesive team, providing necessary resources and motivation, and fostering skill development for team members' growth and career opportunities.
- **Release Staff:** Transition team members to their next assignments at project completion, ensuring recognition of their contributions and preservation of project knowledge.

Define Staff Requirements

- Staff requirements should be defined once project goals, assumptions, and boundaries are clear.
- Two types of skills are identified:
 - Hard skills: Teachable, measurable technical abilities (e.g., programming languages, statistical methods)

- Soft skills: Difficult-to-teach personal traits (e.g., interpersonal skills, empathy, creativity)
- A skill requirements matrix is recommended to systematically define needed skills:
 - Lists project activities
 - Specifies required hard and soft skills for each activity
- Key points about skill development:
 - Soft skills take longer to develop
 - Hard skills can be taught more quickly
 - Some projects may allow time for training to acquire specific skills

Table 5.1 An Example of a Skill Matrix

<i>Activity</i>	<i>Skills</i>
Requirements	Good know-how of the automotive domain. Experience in questionnaire management.
Design	Experience in the use of the IBM Rational suite of modeling tools.
Implementation	Very good know-how of the C++ and WxWidget GUI environment. Experience with test-driven development.

Select Staff:

- **Internal Staff Selection:**
 - Look for personnel within the organization first
 - Consider organizational policies that may limit resource availability
 - Recognize that experienced resources may only be available part-time
 - Consider project timing and priority
 - Use negotiation skills to secure the right people
- **External Staff Selection:**
 - Hire when internal selection doesn't yield required skills or availability
 - Steps in the hiring process:
 - a. Define job description
 - b. Advertise the position
 - c. Wait for applications
 - d. Analyze résumés
 - e. Interview candidates
 - f. Select personnel to hire
- **Job Description and Advertising:**
 - Describe context, company info, required skills, and salary range

- Use professional associations, websites, and personal networks to advertise
- **Résumé Analysis and Interviews:**
 - Review résumés, reference letters, and motivation letters
 - Conduct interviews to assess technical skills and problem-solving abilities
 - Consider work experience and education
 - May use structured multi-stage interview processes in larger organizations
- **Final Selection:**
 - Aim to get a clear picture of both hard and soft skills of candidates

Managing Staff in Software Project Management

Managing staff in software project management focuses on effectively coordinating the team to achieve project objectives while fostering individual growth. This involves:

- **Team Formation**
 - Assemble a cohesive team by integrating individuals with complementary skills.
 - Establish roles and responsibilities to ensure clarity and accountability.
- **Motivating and Supporting the Team**
 - Provide necessary resources, including tools, technologies, and training, to enable efficient work.
 - Motivate the team by recognizing achievements, offering rewards, and maintaining a positive work environment.
 - Address individual concerns to enhance job satisfaction and commitment.
- **Skill Development and Knowledge Sharing**
 - Facilitate learning opportunities to enhance team members' technical and soft skills.
 - Encourage knowledge sharing through collaboration and documentation to benefit both the project and individual career paths.
- **Monitoring Performance and Resolving Conflicts**
 - Use performance metrics and regular feedback to monitor progress and identify improvement areas.
 - Resolve conflicts promptly to maintain a harmonious and productive team environment.
- **Adaptability in Managing Staff**

- Adaptability refers to the ability to adjust task assignments and team structures based on changing project needs or unexpected challenges. For example:
- **Flexible Task Allocation:** Reassign tasks to team members if priorities shift or if someone is unavailable.
- **Handling Challenges:** Address issues like staff leaving the project (staff turnover) by quickly finding replacements or redistributing work among existing team members.

Project Organizational Structures

1. Functional Structure

- Projects are managed by divisional managers who also handle their regular roles.
- Project managers have limited authority and must get approval to access resources from other departments.
- Best for organizations focused on ongoing services rather than project deliverables.
- Suitable for smaller projects or those requiring expertise from specific departments.

2. Project-Oriented Structure

- Projects are the core of the business, with project managers having full authority and resources readily available.
- Suitable for organizations like construction firms, where most work is project-based.
- Effective for large, complex, or high-priority projects.

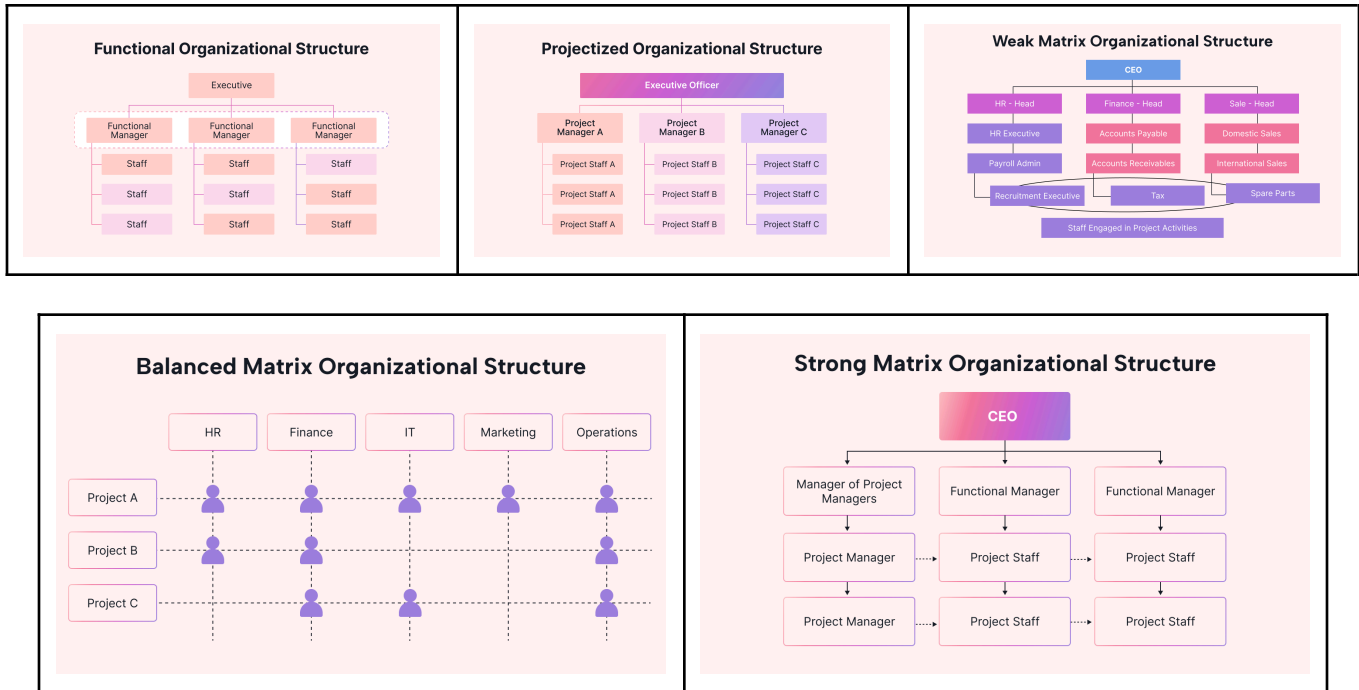
3. Matrix Structure

- Combines functional and project-oriented structures, with project managers having varying authority levels.
- Ideal for projects requiring input from various departments or when a balance between project goals and functional expertise is needed.
- **Weak Matrix:** A weak matrix organizational structure gives functional managers more power than project managers.
- **Strong Matrix:** A strong matrix organizational structure gives more power to the project manager than the functional manager.
- **A balanced matrix** organizational structure gives project and functional managers equal power

4. Composite Structure

- Combines both functional and project-oriented structures, with each reporting to a central executive.

- Used when both structures are needed, like in a government agency managing both maintenance and capital projects.



Managing Communication in Project Management

Effective communication management ensures timely and accurate information reaches the right stakeholders. The types of information exchanged in a project include:

1. **Technical Information:** Essential for carrying out tasks, but must be shared with a common understanding among stakeholders.
2. **Project Status Information:** Provides updates on the project's progress and identifies deviations for corrective action.
3. **Project Decisions:** Helps ensure the project stays on track.
4. **Project Action Items:** Ensures activities are initiated and outputs are collected.
5. **Project Advertisement:** Involves keeping stakeholders engaged, through internal meetings or public dissemination.

5.3.1 Planning a Communication Strategy

Communication often encounters "noise," where the intended message may not align with the recipient's perception. Factors like cultural differences, language barriers, personality traits, and communication mediums all influence this. A communication plan should provide the right information to the right people at the right time, using appropriate channels (e.g., emails, meetings, workshops). Key components of the plan include:

- Information to be exchanged
- Stakeholder roles and communication lines
- Chosen communication means
- Timing and triggers for communication

5.3.2 Communication Styles

Understanding communication styles improves interactions. Common styles include:

- **Aggressive:** Expresses opinions without considering others' feelings.
- **Passive:** Avoids expressing opinions, often to one's detriment.
- **Assertive:** Clearly expresses opinions without disregarding others.

5.3.3 Meetings

Meetings are vital but can become inefficient. Effective meetings should have:

- Clear goals
- Defined participants
- A structured agenda
- Defined timing
- A moderator to keep the meeting on track

Types of Meetings:

1. **Kick-off Meetings:** Start the project or significant tasks, providing essential context and expectations.
2. **Decision-Taking Meetings:** Focus on making informed decisions, requiring proper framing and open discussions.
3. **Audit and Review Meetings:** Assess project status or deliverables, with auditors evaluating progress.
4. **Brainstorming:** Focuses on generating ideas with rules that promote quantity, creativity, and combining ideas.